

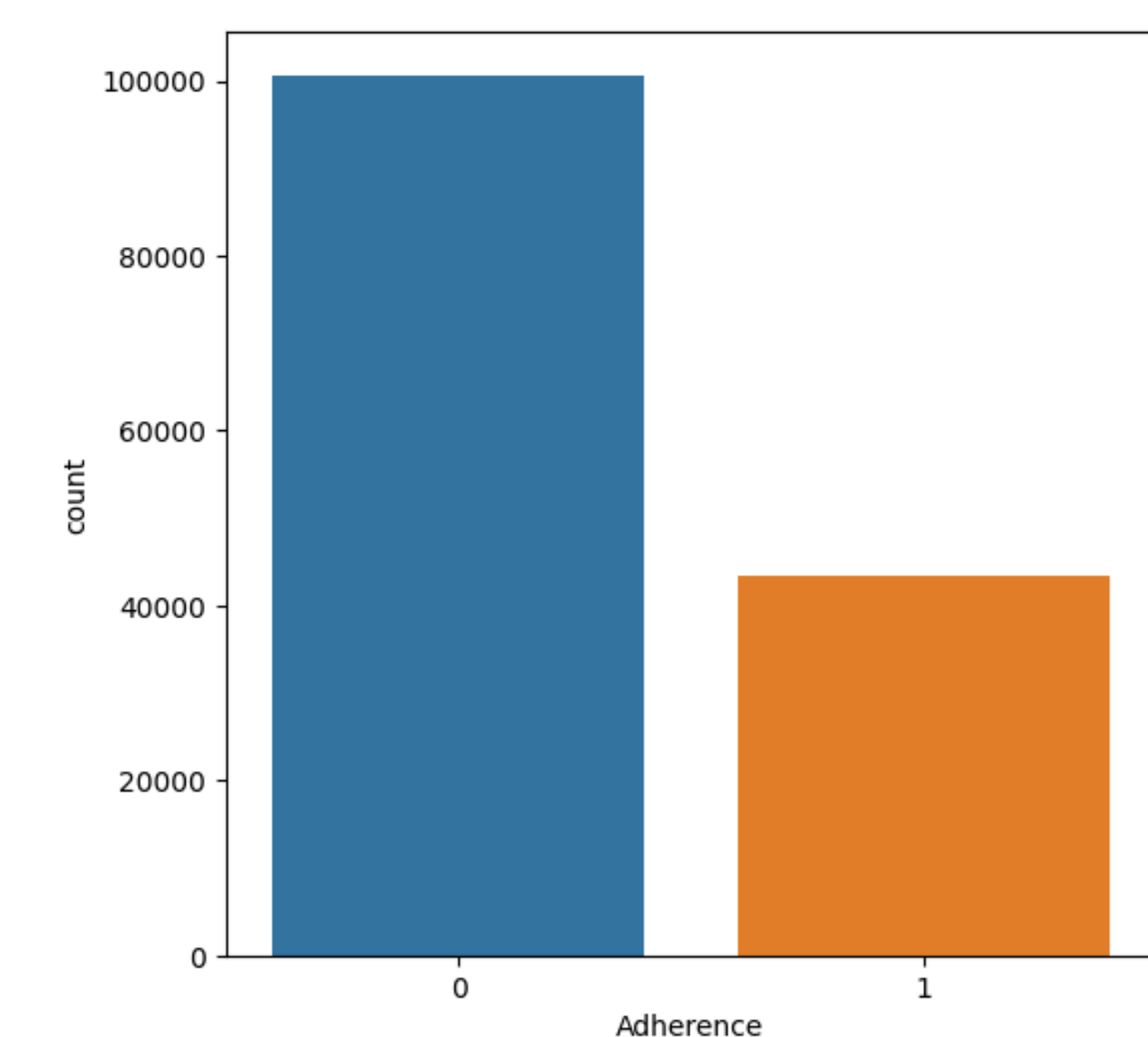
## Introduction

This project investigates the use of machine learning to predict medication adherence in patients. Medication adherence is critical for successful treatment outcomes, and this research aims to identify factors that influence a patient's likelihood of following their prescription regimen.

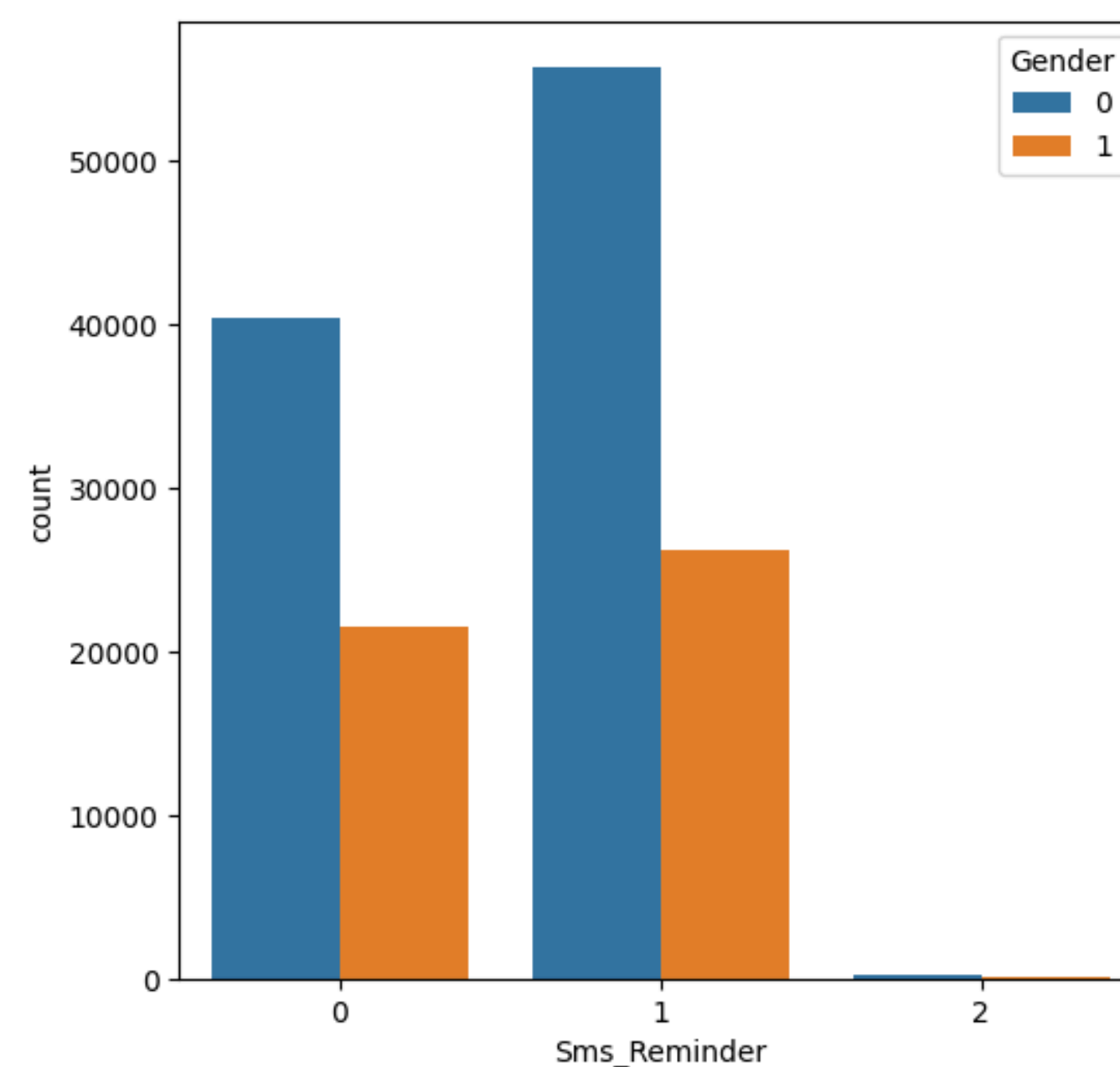
## Data Description

patient\_id: Unique ID of each patient. Age: Age of patient (in years). Gender: Gender of the patient (F- Female, M- Male). Prescription\_period: The number of days, the patient has to follow the prescription. Diabetes: Does the patient have diabetes or not (0: No, 1: Yes)? Alcoholism: Is the patient alcoholic (0: No, 1: Yes). Hypertension: If the patient has hypertension (0: No, 1: Yes). Smokes: Does the patient have an addiction to smoking (0: No, 1: Yes)? Tuberculosis: Does the patient have TB (0: No, 1: Yes)? Sms\_Remainder: The number of remaining SMS sent to patients per day. Adherence: It is a targeted variable. We have to find out if patients are following the prescription or not.

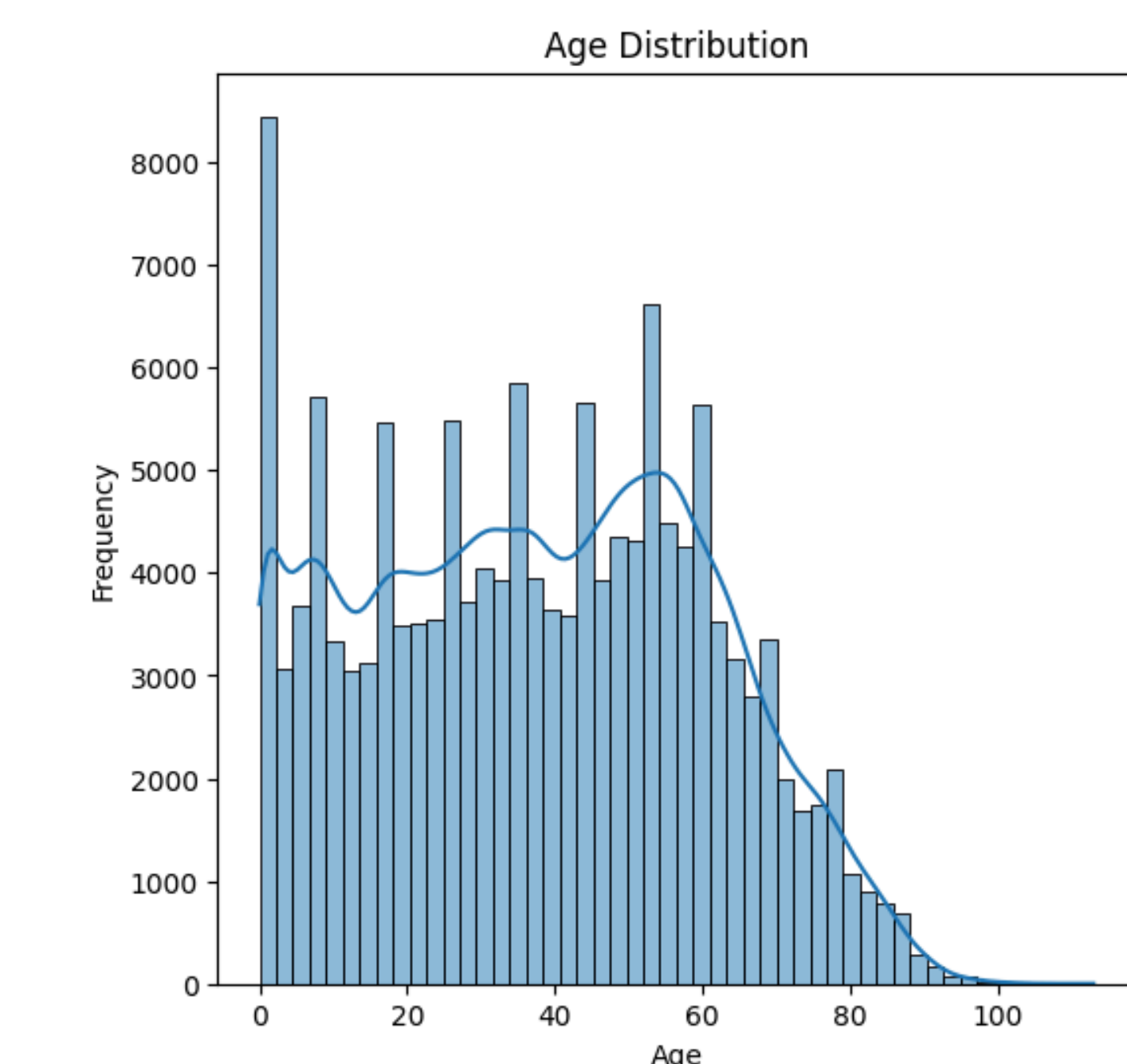
## Exploratory Data Analysis



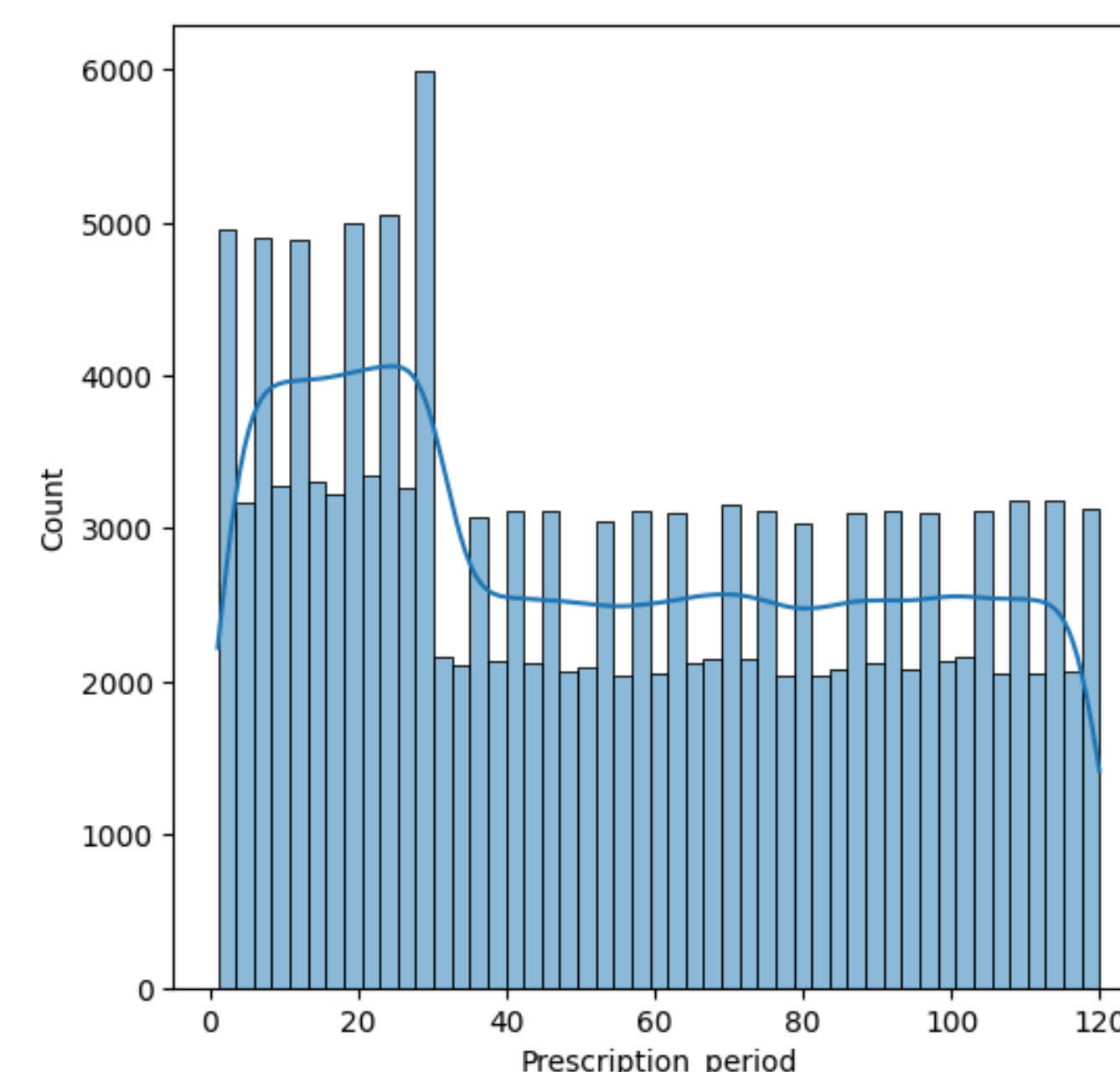
**Figure 1: Adherence Counts:** Distribution between adherent and non-adherent patients.



**Figure 2: SMS Reminders by Gender:** Females receive more SMS reminders than males.



**Figure 3: Distribution of Age:** Ages are well-distributed.



**Figure 4: Prescription Period:** Most prescriptions are for shorter periods.

## Data Transformation and Preparation

- Encoding: Converted categorical variables into numerical values.
- Feature Selection: Dropped the non-important patient\_id column.
- VIF Calculation: Ensured no significant multicollinearity among features.

|   | VIF Factor | features            |
|---|------------|---------------------|
| 0 | 3.5        | Age                 |
| 1 | 1.4        | Gender              |
| 2 | 2.5        | Prescription_period |
| 3 | 1.3        | Diabetes            |
| 4 | 1.2        | Alcoholism          |

|   | VIF Factor | features     |
|---|------------|--------------|
| 4 | 1.2        | Alcoholism   |
| 5 | 1.9        | HyperTension |
| 6 | 1.3        | Smokes       |
| 7 | 1.0        | Tuberculosis |
| 8 | 1.9        | Sms_Reminder |

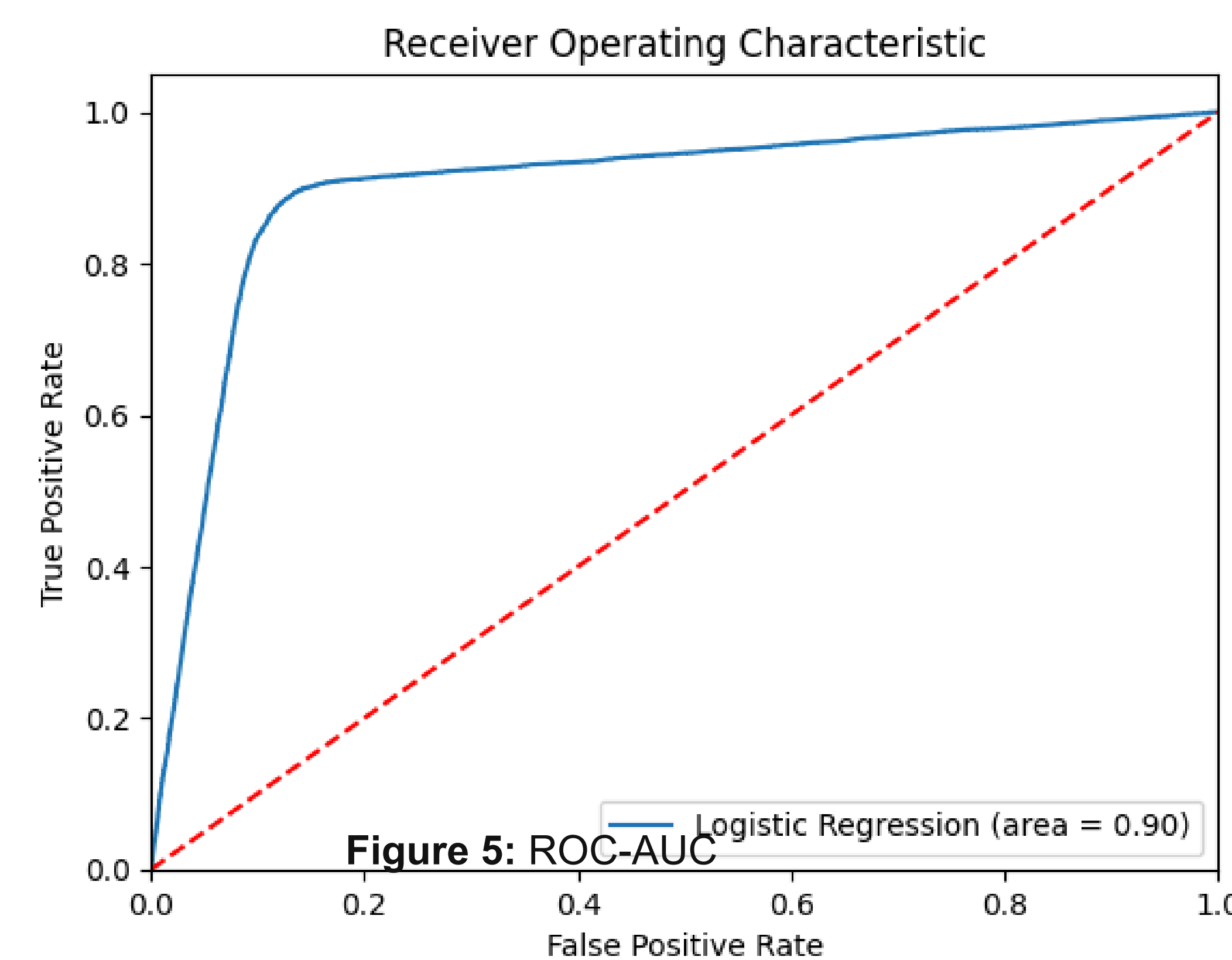
By calculating the VIF, I verified that multicollinearity is not a significant issue in my dataset. This means that the estimates of the regression coefficients will be reliable, and I can proceed with my model without needing to address multicollinearity concerns. This step ensures the robustness and interpretability of regression analysis in predicting patient adherence to prescriptions.

## Model Building

- Train-Test Split: Split data into training (70%) and testing (30%) sets.
- Feature Scaling: Applied standard scaling to numerical features.

## Logistic Regression Model

- Training: Logistic Regression model.
- Evaluation:
  - ROC-AUC Score: 0.88



**Figure 5: ROC-AUC**

- Accuracy: 87.98%
- Confusion Matrix:
  - True Positives: 10721
  - True Negatives: 27333
  - False Positives: 2924
  - False Negatives: 2273

## Model Comparison and Cross-Validation

| Model                    | Mean Test Score | Standard Deviation |
|--------------------------|-----------------|--------------------|
| Logistic Regression      | 0.8808          | 0.0020             |
| SVC                      | 0.8850          | 0.0021             |
| KNN                      | 0.8664          | 0.0005             |
| Decision Tree            | 0.8401          | 0.0022             |
| Random Forest            | 0.8658          | 0.0016             |
| <b>Gradient Boosting</b> | <b>0.8950</b>   | <b>0.0011</b>      |

**Conclusion:** The Gradient Boosting Classifier had the highest accuracy, making it the best choice for this task.

## Conclusion

Here are some key notes we need to consider:

- *Highest Accuracy:* Gradient Boosting Classifier is the best performer with the highest mean test score.
- *Interpretability:* Logistic Regression offers more interpretability and could be preferred where understanding model decisions is crucial.

## Future Work

My next step is to deploy the model to a web application for real-world use. Moreover, I will also develop a website to predict medication adherence based on user input.

## References

- Sabaté, E. et al. Adherence to Long-Term Therapies: Evidence for Action (World Health Organization, 2003).
- Cutler, D. M. & Everett, W. Thinking outside the pillbox? Medication adherence is a priority for health care reform. N. Engl. J. Med. 362, 1553–1555 (2010).
- Miura, T. et al. Effect of digoxin noncompliance on hospitalization and mortality in patients with heart failure in long-term therapy: A prospective cohort study. Eur. J. Clin. Pharmacol. 57, 77–83 (2001).
- Complete report: ([LINK](#))